

Worksheet: Changing the Order of Integration

Double Integrals — Practice Problems

Course 21152 — Infinitesimal Calculus 2, HIT

Procedure.

1. Sketch the domain D described by the given limits.
2. Identify all boundary curves and their intersection points.
3. Fix the *new* outer variable and read its full range from the sketch.
4. For each fixed value of the outer variable, find the range of the inner variable from the boundary curves (express as functions of the outer variable).
5. Rewrite the integral.

Part A Change the Order of Integration

Source: Reznik, Problems in Higher Mathematics, pp. 95–96.

Rewrite each integral as an equivalent iterated integral with the **opposite order of integration**. Note: the result may require splitting into two or three sub-integrals.

A1.

$$\int_0^4 \int_y^{10-y} f(x, y) dx dy$$

Domain bounded by $y = x$ and $y = 10 - x$, with $y \geq 0$.

A2.

$$\int_1^3 \int_{x/3}^{2x} f(x, y) dy dx$$

Domain bounded by $y = 2x$ and $y = x/3$, between $x = 1$ and $x = 3$.

A3.

$$\int_0^3 \int_0^{\sqrt{25-x^2}} f(x, y) dy dx$$

Domain: region inside the circle $x^2 + y^2 = 25$, first quadrant, with $x \leq 3$.

A4.

$$\int_{-1}^2 \int_{x^2}^{x+2} f(x, y) dy dx$$

Domain bounded by the parabola $y = x^2$ and the line $y = x + 2$.

A5. ($a > 0$)

$$\int_0^{2a} \int_{\sqrt{2ax-x^2}}^{\sqrt{4ax}} f(x, y) dy dx$$

Hint: $y = \sqrt{2ax - x^2}$ rewrites as $(x - a)^2 + y^2 = a^2$ (upper semicircle of radius a centred at $(a, 0)$); $y = \sqrt{4ax}$ is a parabola with vertex at the origin.

A6.

$$\int_0^1 \int_{-\sqrt{1-y^2}}^{1-y} f(x, y) dx dy$$

Domain: inside the unit circle $x^2 + y^2 = 1$ to the left of the line $x = 1 - y$, with $y \geq 0$.

A7.

$$\int_0^1 \int_{y^2/2}^{\sqrt{3-y^2}} f(x, y) dx dy$$

Domain: to the right of $x = y^2/2$ (parabola) and inside the circle $x^2 + y^2 = 3$, with $y \geq 0$.

Part B Reverse the Order to Evaluate

Source: Hughes-Hallett et al., Calculus: Single and Multivariable, 6th ed., §16.2, problems 33–37.

Each integral **cannot** be evaluated as written (the inner antiderivative has no closed form). Reverse the order of integration, then evaluate.

B1.

$$\int_0^1 \int_y^1 e^{x^2} dx dy$$

B2.

$$\int_0^1 \int_y^1 \sin(x^2) dx dy$$

B3.

$$\int_0^1 \int_{\sqrt{y}}^1 \sqrt{2+x^3} dx dy$$

B4.

$$\int_0^3 \int_{y^2}^9 y \sin(x^2) dx dy$$

B5.

$$\int_0^1 \int_{e^y}^e \frac{x}{\ln x} dx dy$$

Part C Additional Problems

Standard problems — reverse the order and evaluate.

C1.

$$\int_0^1 \int_x^1 \sin(y^2) dy dx$$

Hint: after reversing, the inner integral reduces to a single factor.

C2.

$$\int_0^1 \int_{\sqrt{x}}^1 e^{y^3} dy dx$$

C3.

$$\int_0^\pi \int_y^\pi \frac{\sin x}{x} dx dy$$

Classic problem — the reversed form evaluates to a familiar integral.